

Monitoring System (Capstone 1st Place)

<https://github.com/pulkitm21/SHM-Capstone.git>

Low-cost vibration monitoring & IoT networking

OBJECTIVE

Design and validate a low-cost structural health monitoring prototype for wind turbines in seismic-prone regions, integrating embedded MEMS sensing, wired network transmission, local data storage, and secure remote dashboard access for continuous vibration monitoring.

MY CONTRIBUTION: FIRMWARE DEVELOPMENT

Sensor drivers: implemented SPI drivers for ADXL355Z (accel.) & SCL3300 (incl.) and I²C driver for ADT7420 (temp.) in ESP-IDF from register-level documentation

Real-time acquisition pipeline: designed timer/ISR-based sampling with lock-free ring buffers to separate deterministic sensor acquisition from FreeRTOS processing and network transmission

Ethernet & PoE bring-up: configured ESP32 Ethernet stack over PoE using LwIP; handled MAC/PHY initialization and network interface management in ESP-IDF

MQTT + fault handling: implemented MQTT client for structured sensor data, status, and fault publishing; added reconnect logic, dropped-sample logging, and NaN handling for partial sensor failures

OUTCOME

Delivered a functional end-to-end prototype from sensor acquisition to Raspberry Pi storage and remote dashboard visualization. Validated simultaneous 7-channel acquisition, fault detection and propagation to the UI, and 24-hour operation with 99.83% data integrity.

